

Optimizing Algorithms for Cancer Signal Detection in Large Genomic Datasets

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BACKGROUND

Early cancer detection improves survival rates, as early diagnoses allow for more effective, timely, and targeted treatments. However, methods for detecting cancer signals in the genome are **computationally intensive**, requiring significant memory and power to analyze millions of DNA reads. This is especially challenging when comparing large datasets or identifying mutation-driven signals, such as those linked to the KRAS oncogene.



Our project addresses this by using **memory-efficient** probabilistic data structures. We represent each genomic sample as a **Bloom filter**, which encodes the presence of **k-mers** while filtering out noise from singleton reads. These lightweight representations allow us to apply fast, approximate **Jaccard similarity** measures to compare samples without processing the full raw data. Ultimately, we aim to build a **lightweight cancer classifier** that allows for **fast detection of cancer signals**.

Category	Traditional Method	Our Method
Data Processing	Full read alignment	Bloom filter representation
Memory Usage	High	Low
Computation Speed	Slow	Fast, approximate
Noise Handling	Limited filtering	Singleton reads filtered out
Similarity Comparison	Full sequence comparison	Jaccard Similarity Comparison
Scalability	Poor with large datasets	Highly scalable

DATA

Obtained RNA-seq data from normal and tumor kidney tissues from the [SRA Project SRP073607](#), as well as gene sequences from the [NCBI database](#).

How We Used the Project

- We used the labeled cancer and normal tissue datasets from this study to **build Bloom filters** for our cancer classification model

Included Tissues

- Sampled across **liver, kidney, intestine, colon**
- 8 normal** tissue samples, **7 tumor** tissue samples

Handling Large File Sizes

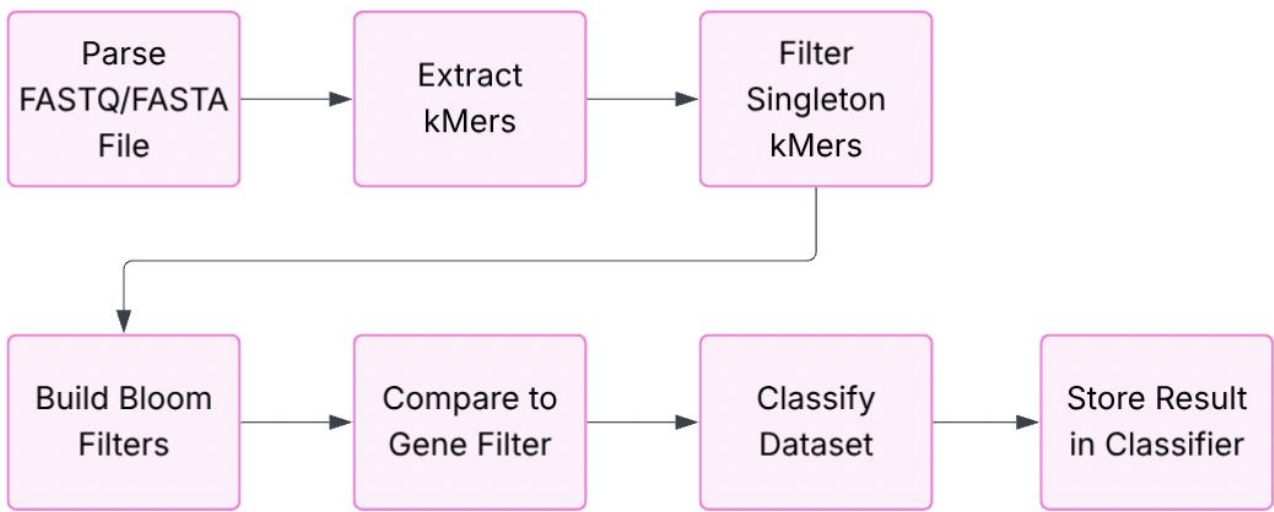
- Each normal / tumor tissue FASTA file ranged from **2GB- 11GB** and we used two FASTQ files of **90 GB** → summing to **210GB - 362GB**
- We **compressed** each of these FASTA files into **200MB** Bloom Filters and **removed singleton noise** reducing overall data size to **3.4GB**

Gene - Level Bloom Filters

- Built a Bloom filter library of cancer-related genes (e.g., **KRAS, TP53, EGFR, BRCA2, MYC**) by converting each gene's FASTA file into a compact filter for lightweight comparisons with tissue.

METHODS

Data Pipeline



We began with raw genomic sequencing FASTA/FASTQ files. Each file was parsed to extract the DNA sequences, which were then broken into overlapping k-mers of length 20. To reduce noise, we filtered out k-mers that appeared only once. This process preserved high-confidence sequences. The remaining k-mers were hashed and inserted into a Bloom filter, producing a compact, probabilistic representation of the dataset.

Each sample's Bloom filter was compared to gene-specific Bloom filters (e.g., KRAS) using Jaccard similarity and k-min hash similarity. These comparisons quantified how closely the sample matched known cancer-associated patterns.

Jaccard Similarity

After encoding each dataset into a bloom filter, we compared them using **Jaccard similarity**, a measure of the overlap between the filters.

A high overlap means the datasets share many k-mers, which could indicate the presence of cancer-related sequences.

We created a Jaccard similarity script to compare the bloom filters from various cancerous and normal cells.

Bloom Filter 1

0	0	1	0	1	0	0
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Bloom Filter 2

0	1	1	0	1	0	1
---	---	---	---	---	---	---

Intersections: **4**

Union: **6**

Jaccard: Intersection / Union
= 4 / 6 = 0.66

PRELIMINARY RESULTS

We started by evaluating various hash functions such as the built-in hash function, HashLib, MurmurHash3, and ultimately chose MurmurHash3 for its proven accuracy, speed, consistency, and deterministic quality.

Using data from the test_hash script, we estimated 45,635 unique k-mers with a length of 20 each, in a real cancer FASTA dataset using a cardinality estimation, which we then used to calculate the false positive rate since bloom filters are probabilistic. In our model, we selected a 200,000,000 bit bloom filter and used 1 hash function.

Unknown1:

Jaccard w/ Tumor	0.4691
Jaccard w/ Normal	0.4484

Classified as: CANCEROUS ✓

Unknown2:

Jaccard w/ Tumor	0.5167
Jaccard w/ Normal	0.5195

Classified as: NORMAL ✓

To further improve accuracy, we removed singletons from the raw data. By removing them, we effectively reduce noise in the data, so that our dataset consists primarily of k-mers with higher confidence, improving the overall reliability and accuracy of our Bloom filter.

We tested our Jaccard similarity script on several different samples, accurately classifying cancer and normal cells.

FUTURE WORK



Broaden Dataset

We plan to **expand our database** by including

- Lung
- Breast

- Pancreatic,
- Colorectal

tissue data, allowing for a more robust and generalizable classifier **across cancer types**.



Cancer Identification

We want to go beyond detection and use our method to **identify** which **specific cancer type** is present in a DNA sample. This allows for more specified cancer classification which would be useful for **targeted treatment**.

REFERENCES

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10814044/>
- <https://www.ncbi.nlm.nih.gov/sra?term=SRP073607>
- <https://www.ncbi.nlm.nih.gov/sra/SRX24361371|accn>



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